

PIERCE COUNTY AP, WA, USA

WMO: 720388

Lat: 47.104N

Lon: 122.287W

Elev: 539

StdP: 14.41

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00469

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.5	24.7	13.5	11.2	25.2	19.1	14.7	28.3	16.1	49.4	13.4	48.3	0.0	10	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.1	88.4	66.8	84.2	65.3	81.1	64.0	68.3	83.4	66.6	80.4	65.2	77.6	3.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	89.4	69.7	62.8	87.3	69.1	61.1	82.0	67.6	32.7	83.6	31.4	80.3	30.3	77.1	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.5	8.8	DB	13.6	95.2	3.6	4.9	11.0	98.7	8.9	101.6	6.9	104.3	4.3	107.9	(4)
(5)			WB	13.2	71.4	3.6	1.6	10.6	72.6	8.5	73.5	6.5	74.4	3.9	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	40.5	41.5	44.9	49.0	55.4	60.0	65.5	66.2	60.1	51.0	44.7	39.3
	DBStd	10.81	6.32	6.09	4.86	4.78	5.30	5.17	4.63	4.48	4.97	4.74	6.64	6.87
	HDD50	1355	298	242	171	73	9	0	0	0	1	45	180	336
	HDD65	5080	761	659	623	481	299	168	48	39	161	435	608	798
	CDD50	1924	3	3	12	42	178	301	480	501	303	75	22	4
	CDD65	172	0	0	0	0	2	19	63	75	13	0	0	0
	CDH74	2837	0	0	3	28	147	339	1034	1052	233	1	0	0
	CDH80	952	0	0	0	5	28	114	367	382	56	0	0	0
Wind	WSAvg	2.0	1.9	2.3	2.8	2.7	1.9	1.9	1.5	1.5	1.3	1.4	2.2	2.1
Precipitation	PrecAvg	41.8	5.9	4.0	4.5	3.4	2.4	1.9	0.7	1.0	1.7	4.0	6.5	5.8
	PrecMax	56.8	12.3	8.6	9.0	7.4	4.3	3.9	1.9	3.7	7.8	9.6	14.4	11.5
	PrecMin	31.2	2.4	0.2	1.5	1.0	0.3	0.3	0.0	0.0	0.1	0.7	1.7	1.2
	PrecStd	6.7	2.3	2.0	1.7	1.6	1.1	0.9	0.5	0.9	1.6	2.0	3.1	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	61.0	70.4	78.6	83.7	88.5	93.2	92.6	86.3	71.5	64.4	59.1
		MCWB	51.3	50.7	54.8	60.0	61.2	67.6	66.9	66.8	66.1	59.5	57.6	52.3
	2%	DB	54.6	56.6	62.8	69.9	78.7	83.6	88.4	89.5	80.7	66.0	60.8	54.4
		MCWB	49.9	49.6	51.3	56.2	61.0	65.9	66.2	67.2	64.1	57.2	55.0	50.3
	5%	DB	52.3	54.1	57.2	63.6	73.0	77.0	83.9	84.0	75.3	63.2	57.0	51.8
		MCWB	49.0	48.5	49.4	51.9	59.1	62.0	65.2	65.9	62.3	55.4	53.1	49.1
	10%	DB	49.7	51.6	54.4	58.9	68.0	72.5	80.6	80.8	71.9	60.7	54.5	48.2
		MCWB	47.7	48.4	48.1	49.9	57.7	60.3	63.9	64.3	60.7	55.3	51.5	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	54.7	56.1	60.9	65.4	69.3	71.4	70.0	68.7	62.7	59.8	54.8
		MCDB	56.4	57.7	61.1	75.7	79.5	85.2	89.1	86.9	81.4	66.0	63.8	56.6
	2%	WB	51.7	52.2	53.2	56.6	62.6	66.5	68.0	68.1	65.9	59.6	56.5	52.1
		MCDB	53.3	54.4	59.2	66.7	74.4	79.5	83.0	83.7	76.2	62.8	59.4	54.1
	5%	WB	49.4	50.1	51.3	53.9	60.2	63.5	66.1	66.4	64.0	57.5	54.2	48.5
		MCDB	50.9	52.1	56.3	60.3	70.9	73.1	80.3	80.7	70.7	60.7	56.6	50.0
	10%	WB	47.1	47.8	49.4	51.7	58.1	61.5	64.5	65.1	62.4	55.4	52.2	46.3
		MCDB	48.5	50.0	53.8	56.9	66.5	71.5	77.3	77.8	68.3	58.9	54.2	47.5
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.3	17.2	18.0	20.3	20.3	24.9	24.1	20.8	17.6	14.6	11.9
		MCDBR	15.6	20.0	23.7	28.6	31.7	30.8	34.1	33.1	29.4	22.3	16.8	13.3
		MCWBR	10.7	12.8	13.7	14.8	15.4	13.4	12.9	12.6	13.9	13.9	11.5	9.2
	5% WB	MCDBR	12.5	14.4	19.9	23.7	27.0	27.3	29.3	29.2	23.7	17.7	14.9	11.7
MCWBR		9.7	10.6	12.6	13.2	13.9	12.6	12.4	12.0	12.2	12.5	11.1	9.2	
Clear-Sky Solar Irradiance	taub	0.319	0.315	0.323	0.358	0.374	0.367	0.349	0.356	0.344	0.327	0.319	0.328	
	taud	2.465	2.500	2.514	2.398	2.379	2.424	2.493	2.480	2.509	2.561	2.521	2.373	
	Ebn at Noon	244	273	288	285	282	285	288	282	276	265	243	221	
	Edh at Noon	21	25	28	35	37	36	33	32	28	23	20	21	
All-Sky Solar Radiation	RadAvg	333	610	911	1338	1682	1856	2072	1767	1296	742	392	272	
	RadStd	49	81	105	121	138	174	148	120	113	83	50	40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page